

# CocoConnect – Refined Stakeholder Requirements Document

## 1. Introduction

CocoConnect is a digital ecosystem designed to connect coconut estate owners, buyers, workers, service providers, and administrators through a single platform. The system aims to address challenges related to product selling, workforce management, harvest recording, trust between stakeholders, and estate management.

Requirements were gathered through discussions with industry stakeholders from the Coconut Growers Association of Sri Lanka and estate owners.

## 2. Stakeholders

### Primary Stakeholders

- Estate Owners
- Estate Managers
- Buyers
- Sellers
- Coconut Pluckers
- Huskers
- Fertilization Workers
- Coconut Gathering Workers
- Tractor Suppliers
- Equipment Suppliers
- Platform Administrators

### Secondary Stakeholders

- Future agricultural product sellers
- Tourists and tourism operators
- Logistics providers

## 3. Current Business Process Issues

### Product Selling Issues

- No centralized marketplace for coconut products.
- Buyers and sellers struggle to identify trustworthy trading partners.
- Product prices are negotiated manually.
- Limited visibility of available products and quantities.

### Workforce Issues

- Difficulty finding skilled workers when needed.
- No platform to verify worker experience or reliability.
- No rating system for workers.

### Harvest Management Issues

- Harvest records are maintained manually using Google Sheets.
- Harvest data entry is time-consuming.
- Lack of historical analytics and reporting.

### Trust Issues

- Trust concerns between buyers and sellers.
- Trust concerns when hiring workers.
- Lack of reputation and review mechanisms.

## 4. Functional Requirements

### Module 1 – User Management

FR-01: Users shall be able to register accounts.

FR-02: Users shall be able to log in securely.

FR-03: Users shall be assigned role-based access.

Roles include:

- Estate Owner
- Estate Manager
- Buyer
- Seller

- Worker
- Service Provider
- Administrator

FR-04: Users shall be able to update profile information.

FR-05: Administrators shall verify user accounts.

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## **Module 2 – Product Marketplace**

FR-06: Sellers shall be able to create product listings.

FR-07: Buyers shall be able to browse available products.

FR-08: Buyers shall be able to search and filter products.

FR-09: Sellers shall be able to manage inventory quantities.

FR-10: Buyers shall be able to contact sellers.

FR-11: The system shall support multiple selling methods:

- Per kilogram
- Per nut

FR-12: The system shall support product categories:

- DC Coconut
- Curry Coconut
- Coconut Husk

FR-13: Sellers shall be able to specify:

- Price
- Quantity
- Location
- Product type
- Selling method

FR-14: Buyers shall be able to view seller ratings.

## **Module 3 – Workforce Marketplace**

FR-15: Workers shall be able to create service profiles.

FR-16: Estate owners shall be able to search workers.

FR-17: Workers shall be categorized by service type.

Supported services:

- Coconut Plucking
- Husking
- Fertilizing
- Coconut Gathering
- Coconut Loading
- Tractor Services
- Equipment Supply

FR-18: Workers shall be able to specify charging methods.

Examples:

- Per Tree
- Per Day

FR-19: Estate owners shall be able to hire workers through the platform.

FR-20: Workers shall receive job requests.

FR-21: Workers shall be rated after completing jobs.

FR-22: Estate owners shall be rated by workers.

Workers will be rated by Managers based on their performance.

## **Module 4 – Harvest Management**

FR-23: Estate managers shall record harvest details.

FR-24: The system shall store:

- Harvest date
- Tree count
- Coconut count
- Total harvest

FR-25: The system shall maintain historical harvest records.

FR-26: Estate owners shall view harvest reports.

FR-27: The system shall support offline data collection.

FR-28: The system shall synchronize records when connectivity becomes available.

## **Module 5 – Trust & Rating System**

FR-29: Buyers shall rate sellers.

FR-30: Sellers shall rate buyers.

FR-31: Estate owners shall rate workers.

FR-32: Workers shall rate estate owners.

FR-33: Ratings shall contribute to user reputation scores.

FR-34: Reviews shall be publicly visible.

## **Module 6 – Estate Management**

FR-35: Estate owners shall register estates.

FR-36: Estate owners shall maintain estate information.

FR-37: The system shall store:

- Estate location
- Acreage
- Number of trees

FR-38: Estate owners shall track harvest performance.

FR-39: Estate owners shall monitor workforce activities.

## **Module 7 – Administration**

FR-40: Administrators shall verify users.

FR-41: Administrators shall manage reported accounts.

FR-42: Administrators shall monitor platform activity.

FR-43: Administrators shall remove fraudulent listings.

FR-44: Administrators shall manage user permissions.

## 5. Future Enhancements

The following features were identified by stakeholders but are outside the initial project scope.

### Additional Marketplace Products

- Pepper
- Mango
- Lemon
- Orange
- Pineapple
- Poultry Products

### Tourism Services

- Estate Cabana Booking
- Estate Tourism Experiences

### Digital Payments

- Online Payment Gateway
- Escrow-based Payments
- Secure Transaction Handling

### Advanced Analytics

- Yield Prediction
- Productivity Analysis
- Price Trend Analysis

## 6. Non-Functional Requirements

NFR-01: The system shall support mobile devices.

NFR-02: The system shall provide offline functionality for harvest recording.

NFR-03: System response time shall be less than 3 seconds under normal conditions.

NFR-04: User passwords shall be securely encrypted.

NFR-05: The system shall provide role-based authorization.

NFR-06: The system shall maintain 99% service availability.

NFR-07: The system shall support future expansion to additional agricultural industries.

NFR-08: The interface shall support English, Sinhala, and Tamil languages.

## 7. Domain Knowledge Captured During Requirement Gathering

- Two coconut categories currently sold:
  - DC Coconut
  - Curry Coconut
- Selling methods:
  - With Husk
  - Without Husk
- Coconut plucking methods:
  - Pole (Kekka)
  - Tree Climbing
- Common worker charging methods:
  - Per Tree
  - Per Day
- Coconut quality indicators:
  - Average Nut Weight (~615g)
  - Kernel Thickness
  - Estate Location
- Ideal plantation density:
  - 64 trees per acre
- Recommended spacing:
  - 26 feet between coconut trees
- Current payment methods:
  - Bank Transfer
  - Cheque